

WashU Medicine

Note: Lower scores indicate better usability

Figure 2. Prototypes of ML-Augmented Risk Report

User Experiences with Risk Report	Phase I	Phase II	Phase III
Perceived Clinical Value of the Risk Report Supports anticipatory management of postoperative complications Integrates into handoff and documentation workflows Promotes clinician education and practice improvement	- Early identification and prevention of postoperative complications <i>"It's useful for providing anticipatory guidance if someone is high risk for delirium, then you might want to implement additional precautions"</i> [P2] - Primary display key to risk prediction interpretation - Secondary display (top 10 features) enhances engagement <i>"... this is what this risk scoring means, may make me more cognizant of what's happening and pay more attention to that feature"</i> [P3]	- Helpful for handoff preparation <i>"This [risk report] is helpful to me, because then I copy and paste it [information] into my note, and I'm not supposed to start my notes [in the EHR] before the patient comes to unit [ICU], but now I have a leg up on knowing what's going on when they get here and being ahead on my notes."</i> [P12] - More useful as educational tool <i>"Primary display could be helpful during the handoff, but the secondary display of the report could be more of an educational purpose"</i> [P3]	- Enhance anesthesia risk assessment and handoff discussions <i>"I can see anesthesia doing this more than the surgical staff... and somewhere in their report, there is space for it [risk assessment]"</i> [P13] - More beneficial as learning tool for discovering previously unknown relationships <i>"I think it's kind of academic in that sense that it's interesting... it might be a little bit more useful for them [researchers] than others. It just kind of piques your interest, and maybe it can be another tool in the toolbox"</i> [P13]
Challenges in Risk Interpretation Difficulty interpreting risk prediction scores Confusion with consolidated prediction models Limited understanding of model features Challenges interpreting explainability tools	- Risk stratification: difficulty understanding due to absence of actual risk percentages and threshold values, despite color-coding <i>"Ok, the other thing which is a problem is that I don't know what you mean by high risk. There is like no output of what? What is the definition of high risk?"</i> [P6] - Contributing feature values: difficulty interpreting; uncertainty about measurement timing during case <i>"...What's the tidal observed that's like Max current? When's the nitrogen nitric oxide measurement to derive from?"</i> [P6]	- Combined DVT/PE model: difficulty understanding associations between contributing features and DVT/PE <i>"This person, though, I did feel like is at high risk for thromboembolic disease because of the nature of their surgery but not really related to any variables there [contributing features on secondary display]"</i> [P9] - Contributing feature associations: still not comprehensible despite evidence-based explanations, due to features not typically associated with risk complications <i>"... it says there's no impact on the... complication... If it's saying here that cholesterol has no impact on delirium, how is that included in the model?"</i> [P3]	- Shapley (SHAP) values: confusion due to unfamiliarity with concept and interpretation <i>"The what value?... yeah, I don't know what a Shapley value is... I don't think that people would care... the SHAP value doesn't help me, because I don't know how to interpret it"</i> [P8] - Contributing features "not typically associated" wording in hover-over explanations: questioned its presence <i>"Right, so why is this [the contributing feature] here? That's exactly what I'm saying. Like, what does pulse have to do with any of this... and then your little icon says it, so then why is this even here?"</i> [P9]
Content and Visual Design Inconsistencies Inconsistencies between visual displays and underlying data Graphical encodings did not clearly communicate risk Color-coding conflicted with clinical judgment Presentation issues reduced confidence in interpretation	- Risk assessment: discrepancy between graph ("high" risk) and overall risk prediction ("low" risk) <i>"It's a little confusing because this patient was labeled low risk [for pneumonia on primary display]...most of the factors have red bars which suggests this patient should have increased risk [for pneumonia]..."</i> [P1] - Normal values shown as red bars (high risk): led to confusion <i>"It's relatively normal pCO2, so how is that an increasing risk? So, I think just some of the data [contributing features] that's been shown as contributing the highest is confusing"</i> [P3]	- Bar length and corresponding values: lack of correlation led to confusion <i>"I think what's confusing here is that there is a length of the red bar, and the number next to the red bar does not refer to the length of the red bar"</i> [P1] - Secondary display graphs and primary display risk prediction inconsistency <i>"The graph displayed more red bars indicating high risk, although the overall risk prediction was low"</i> [P10]	Color coding of risk complications: sometimes contradicts participants' own initial assessment <i>"I do think delirium should be red. I don't know why 30% delirium is okay with everyone... the yellow makes it feel... implied it's less important or something"</i> [P9]
Unintended Consequences of Risk Report Use Potential for misinterpretation of model outputs Concerns about legal and clinical accountability Uncertainty about how to incorporate the report into clinical workflow Risk of overreliance on or disregard of model predictions	- ML model hallucinations: potential cognitive manipulation of users <i>"I think it's always going to be hard to tell a story when you have a machine learning model that's learning from 200 features. If [ML model] is going to come up with stuff [contributing features] that is correlated [with risk complications] but is not clinically obvious why it should be correlated"</i> [P1] - Associations: risk of misinterpretation as causation, leading to poor clinical decisions <i>"People [clinicians] might make causal relationships and lead them to make bad decisions"</i> [P2]	Clinical decisions differs from model recommendations: may require additional documentation to mitigate risk <i>"If this [risk report] is in the patient record and the patient ends up truly having a PE, and I ignored this, now do I need to go back in the patient record and write all the reasons why I don't think this patient has a PE to cover myself legally? Because this [risk report] is now telling me that machine learning is suggesting PE. In the state of court, is it going to say that [clinician] has seen this many patients and say that he's better than the machine learning?"</i> [P10]	- Uncertainty with workflow changes based on ML risk report <i>"When I look at a patient in the ICU, I am not determining whether they are at high, medium, or low risk for a DVT. That is quaternary to me... I am not necessarily going to treat patients differently a priori because I suspect they are at high risk"</i> [P8] - Lack of clarity on how to apply the information to clinical decision-making <i>"What do you want me... you're making a tool, so you went into all this effort to do it, but what do you want me to do with this information? Like, what is your goal for me to see this and feel what?"</i> [P9]

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graph LR
    EV[External variables] --> PU[Perceived Usefulness]
    EV --> PEU[Perceived Ease of Use]
    PEU --> PU
    PU -- TRUST --> ATU[Attitude Towards Use]
    PEU --> ATU
    ATU --> BI[Behavioral Intention to Use]
  
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External variables

- Risk score quality and feature explainability
- Technical, cognitive, affective usability of risk report
- Workflow integration of risk report
- Clinical relevance of risks and features
- Appropriate clinical thresholds (surgery type, impact) for fewer false positives and fewer false negatives
- Training for adoption
- Technical AI literacy
- Context of use aspect (when, how, what)
- Actionability of risk features

Perceived Usefulness

- Early identification of risks
- Anticipatory management
- Enhanced risk monitoring
- Educational support
- Clinical decision support
- Empower clinicians with insights
- Effective handoffs with Q/A

Perceived Ease of Use

- Easy to understand
- Easy navigation
- Intuitive colors
- Simple interface (with hover-over for on-demand information)

Attitude Towards Use

- Positive experiences interacting with the report
- Agreement with risk scores and features
- Supports a fresh eye perspective
- Reduced cognitive burden to interpret the risks and features

Behavioral Intention to Use

- Improvements in postoperative workflow
- Fewer adverse events and near misses
- Reduced OR call-backs
- Reduced realized errors
- Clinician buy-in and uptake

CLINICAL VALUE ADDED

TRUST

- UCD approach enabled iterative refinement of the ML-augmented risk report, resulting in high usability and perceived feasibility for postoperative care management
- Usability and interpretability challenges identified opportunities to improve risk communication, visual design, and model explainability
- Consistent with prior studies [10], model explainability and interpretability remain key barriers to clinician trust and adoption of AI tools in clinical workflows
- Findings underscore the importance of iterative user-centered design to optimize AI tools for clinical use and facilitate successful implementation
- To inform implementation, 100 postoperative handoffs were observed, and existing handoff workflow was mapped to identify opportunities for seamless integration

The screenshot displays the Axiom web application interface, specifically the 'Results Review' tab. The main content area shows a table of 'Assessment Record' for 'JULIUS S. JULIUS S.' with columns for 'Assessment Record', 'Assessment Date', 'Assessment Type', 'Assessment Status', 'Assessment Score', and 'Assessment Comment'. The table lists two records: one for 'JULIUS S. JULIUS S.' on 10/10/2018 with a score of 100, and another for 'JULIUS S. JULIUS S.' on 10/10/2018 with a score of 100. The right sidebar contains a 'Summary' section with a 'Results Review' tab, a 'Summary' section with a 'Results Review' tab, and a 'Results Review' section with a 'Results Review' tab. A red arrow points to the 'Results Review' section.

Figure 4. Proposed ML-Augmented Risk Report Integration with Epic-embedded Handoff Tool

- Next steps include:
 - Refining the risk report based on Phase IV findings
 - Integrating the ML-augmented risk report into the Epic-embedded postoperative handoff tool using a SMART-on-FHIR link (**Fig. 4**)
 - Evaluating the clinical effectiveness, implementation, and impact of the ML-augmented handoff report on care continuity, clinician adoption, and patient safety outcomes

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